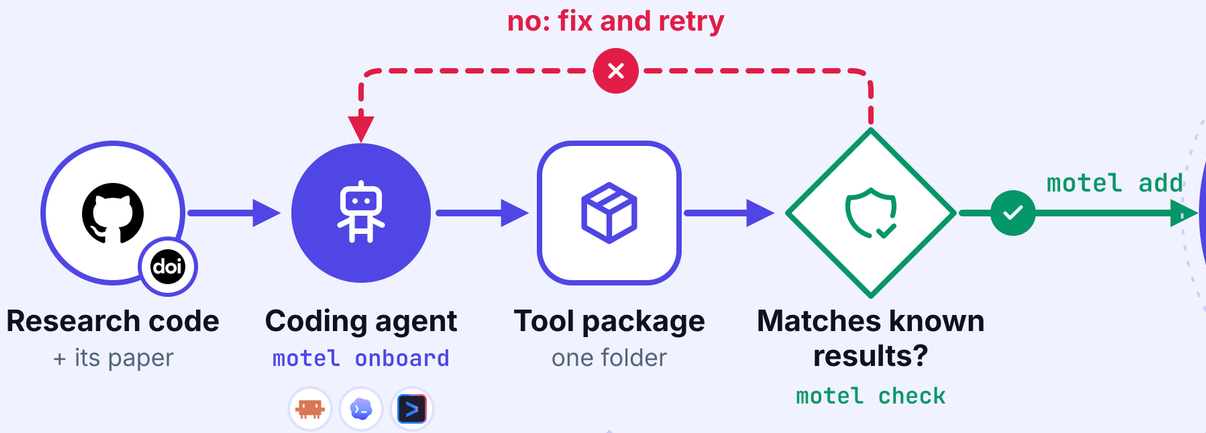
1 BUILD ONCE



Inside a tool package

`motel.yaml`	pinned source + environment	`tests/`	known answers to reproduce
`api.py`	typed functions, with units	`patches/`	fixes to the original code
`SKILL.md`	when and how agents use it		



Motel MCP server
serving 50 verified tools

2 USE EVERYWHERE

MCP

Harness

Is silicon a direct-gap semiconductor?

`find_tools "band gap"`

`run_tool materials-project · Si`

`run_tool gpaw · band structure`

No: **indirect** gap, Γ → X
0.59 eV (PBE; measured 1.12 eV)

Verified · Cited · Reproducible

Claude · Codex · Gemini CLI · Cursor · LangGraph

3 THE CATALOG

Quantum chemistry
- PySCF
- Psi4
- Quantum ESPRESSO
- GPAW
- xTB
- CREST

Molecular simulation
- LAMMPS
- OpenMM
- ASE
- ML potentials
- phonopy
- phono3py
- MDAnalysis
- OVITO

Materials
- pymatgen
- Materials Project
- robocrys
- PyXtal
- surfaces
- ion diffusion
- Zeo++
- RASPA3

Thermodynamics
- Cantera
- CoolProp
- pycalphad
- Reaktoro
- thermo
- FiPy

Spectroscopy
- xraylarch
- xraylib
- pyOpenMS
- MixSense NMR
- AtomBridge

Energy
- PyBaMM
- impedance.py
- pvlib

Chemistry & drugs
- RDKit
- RDKit descriptors
- Open Babel
- AiZynthFinder
- AutoDock Vina
- PROPKA
- UMA-Dock
- peptide-agent

Knowledge & lab
- SKY synthesis
- ARIA doping
- MP analogies
- CLUE
- Catalyze
- AtomicShorts